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Salience, transparency, and self-nudging: a digital nudge to promote healthier food product choices

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ABSTRACT
Information Systems (IS) research has proposed digital nudges as a promising toolset for supporting consumers in making better decisions. However, there is a vivid academic debate regarding the ethical underpinnings of nudging and potential measures to alleviate ethical concerns. One particularly promising approach to digital nudging is leveraging salience effects. Yet, the potential of digital salience nudges remains underexplored in IS literature, and little is known about how measures proposed to alleviate ethical concerns affect their behavioural impact. In a preregistered online experiment ($N = 600$), we assess the effects of a digital salience nudge on the healthiness of food product choices. Additionally, we examine how two proposed ethical measures, namely transparency and self-nudging, influence the nudge's effects. We find that implementing the digital salience nudge in an online supermarket decreased the proportion of unhealthy food products chosen by 63% and increased the proportion of healthy alternatives by 30%. Notably, disclosing transparent information about the nudge did not impact its effectiveness. Offering the nudge as a self-nudge led to comparable effects to a preset nudge when combined with transparent information. We conclude that digital salience nudging is effective and that nudges can be ethically designed without compromising their behavioural impact.

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1. Introduction

Over the past decade, the importance of online sales channels has increased significantly. In 2021, the share of ecommerce turnover of European Union (EU) enterprises was 20%, whereby the proportion of Europeans who reported purchasing products online in the past 12 months increased by almost 70% from 39 in 2011 to 66% (Eurostat, 2021, 2022).

To deal with the vast product assortment offered by ecommerce retailers (Brynjolfsson et al., 2003, 2011), consumers often apply heuristics (i.e., mental shortcuts or rules of thumb) in their decision-making processes (Arnott & Gao, 2022; Benartzi & Lehrer, 2015; Dennis et al., 2020; Mirsch et al., 2017). These heuristics can lead to myopic choices that are not in the individuals' best interests and not aligned with their longterm preferences (Evans, 1989; Goes, 2013; Milkman & Beshears, 2009).

A promising strategy for overcoming these issues and supporting individuals in making more foresighted decisions is the concept of nudging, which Thaler and Sunstein (2008) first introduced. It builds on behavioural economic theory and refers to small changes in the decision structure (i.e., the choice architecture) that influence human behaviour. The term *nudge* refers to "any aspect of the choice architecture that alters people's behaviour in a predictable way without forbidding any options or significantly changing their economic incentives" (Thaler & Sunstein, 2008, p. 6). Nudges help individuals act in their best self-interest and often work by leveraging the same cognitive boundaries and biases that can lead to suboptimal decisions (Hansen, 2016).

IS research has coined the concept of *digital nudging*, which refers to nudges implemented in digital environments (Schneider et al., 2018; Weinmann et al., 2016). While analogue and digital nudges share the same roots, digital nudges can leverage a vast additional spectrum of digital tools, including changes in graphical user interface design elements, user reviews, push notifications, personalised real-time feedback, or reminders (Mirsch et al., 2018; Weinmann et al., 2016). The versatility of digital environments offers multiple advantages for nudges compared to analogue decision contexts. First, digital nudges are much easier to scale because they are usually fast and inexpensive to implement (Mirsch et al., 2017). Second, they can leverage user tracking to implement personalised nudging approaches (Peer et al., 2020; Weinmann et al., 2016). Third, designers such as ecommerce retailers can precisely control and adjust the information flow directed to the end user, which is difficult to realise in analogue choice environments (Dennis et al., 2020).

One set of interventions that particularly benefit from these opportunities are digital salience nudges.

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These leverage the so called salience bias, which states that salient (i.e., prominent or distinctive) stimuli receive more attention from individuals and thus disproportionate weight in a decision (Bordalo et al., 2013; Kahneman et al., 1982). By adjusting the salience of specific aspects in the choice environment, salience nudges can lead to shifts in attention and thus also individual behaviour (Noggle, 2018). While salience nudges are also employed in analogue contexts (see e.g., Cadario & Chandon, 2020; Wilson et al., 2016), digital choice environments offer highly adaptable mechanisms and nearly unlimited options to precisely control and adjust user interfaces. This high level of customisability can be leveraged by implementing particularly targeted, flexible, and optimised salience manipulations that would not be possible in analogue decision contexts (Mirsch et al., 2017; C. Schneider et al., 2018). If employed as digital nudges, salience manipulations could therefore extend the use cases and amplify the effects of conventional salience nudges. Investigating the effects of salience nudges in digital environments is therefore promising for guiding digital choice architects and designers to find additional potentially more effective ways of supporting consumers to act in line with their preferences.

In the past, IS research has examined salience effects in multiple contexts. A large proportion of studies have focused on the effects of highlighting specific information on individual behaviour (Jaeger & Eckhardt, 2021; Kim & Dennis, 2019; Y. Lee & Chen, 2019; Tiefenbeck et al., 2018; Wang et al., 2016; Willems et al., 2019; Yi et al., 2019). Others have investigated the impact of different presentation formats on the perception of banners and advertisements (Ahn et al., 2018; Cheung et al., 2017; Hong et al., 2004, 2021; Sun et al., 2013) or reported occurrences of the salience bias in online environments (Kuem et al., 2020; H. C. B. Lee et al., 2018). Together, these studies highlight that attention is a key driver of human behaviour in the digital world and provide substantial evidence that salient stimuli elicit strong attentional capture effects (Tams et al., 2018). Yet, surprisingly few studies have investigated how these effects could be instrumentalised to support individuals in making decisions that better align with their long-term preferences or protect them from making harmful decisions. While some studies investigate digital salience nudges and suggest their effectiveness (Berger et al., 2020; Franz & Benlian, 2022; Jesse et al., 2021; Kim & Dennis, 2019), they did not focus and elaborate on the mechanisms underlying salience effects and did not embed their approaches in existing theoretical accounts. Therefore, it is unknown how and to what extent digital salience nudges can guide consumer decision-making in digital choice environments. Building on visual and attribute salience theory, we provide a first step to close this gap. We thus contribute to the ongoing discussion at the intersection of behavioural economics, choice architecture, and technology-mediated choices (Adjerid et al., 2018; Goes, 2013; Johnson et al., 2012) about how digital nudges can effectively support consumers in their decision-making. To do so, we introduce a novel digital salience nudge that operates by reducing the visual salience of undesired choice options and investigate the following overarching research question:

RQ_1 How effective is a digital salience nudge in guiding consumer decisions in an online shopping context?

The literature on (digital) nudging is characterised by two overarching themes: on the one hand, the large potential benefits for individuals and society in various application domains, and on the other hand, a controversial discussion about ethical concerns that are inextricably linked with that potential. These concerns arise from the covert nature of many nudges, which can exert influence on behaviour without explicit consent or in some cases even without conscious awareness of the nudged individuals. Therefore, a vivid academic and political discussion has raised multiple key arguments against their implementation and even deemed them unethical. Among the main points of criticism are restrictions on the individual's freedom of choice and autonomy or the common opacity of nudges (Acquisti et al., 2017; Fabian & Pykett, 2022; Hansen & Jespersen, 2013; Schmidt & Engelen, 2020; C. Schneider et al., 2018). In IS research, several articles echo this criticism and provide concepts on how to design ethical digital nudges that might overcome these points of criticism (e.g., Lembcke et al., 2019; Meske & Amojo, 2020a; Renaud & Zimmermann, 2018). However, empirical studies investigating the impact of these proposed measures on the effectiveness of digital nudges are scarce. Therefore, it remains unclear whether employing digital nudges in a more autonomy-preserving and transparent way might jeopardise their behavioural effects.

To close this gap, we examine how two different measures that have been proposed to solve the potential issues associated with digital nudging, namely transparency and self-nudging, impact the effectiveness of our digital salience nudge (Bruns et al., 2018; Kozyreva et al., 2020; Lembcke et al., 2019; Reijula & Hertwig, 2022). Thus, we extend the focus on salience theory as our theoretical framework in research question 1 by contributing to ongoing debates inextricably linked with the potential of digital nudges, regarding the ethical use of information technology and digital nudges (Lembcke et al., 2019; Meske & Amojo, 2020a; Myers, 2021; C. Schneider et al., 2018; Weinmann

et al., 2016) as well as human dignity in digitised environments (Leidner & Tona, 2021) by answering the following research question:

RQ$_2$ To what extent do transparency and self-nudging interventions affect the effectiveness of a digital salience nudge?

We answer our research questions by focusing on one area in which millions of consumers struggle to align their everyday decisions with their long-term preferences, namely food choice. Although many individuals aim to buy healthy food products and increase their diet's nutritional quality, they often fail to put their good intentions into practice and instead indulge in unhealthy, high-calorie products. Salience bias is one of the key drivers of these issues (Ares et al., 2020) as the positive sensations associated with the consumption of unhealthy food products are typically more vivid to consumers than their long-term impact on their health. Because an ever-increasing proportion of food products is purchased via online channels (Verschueren et al., 2021) and decisions in this context are often based on heuristic and attentional strategies (Schulte-Mecklenbeck et al., 2013; Torma et al., 2018), digital salience nudges are a promising approach to support consumers in making healthier online food product choices (Wilson et al., 2016).

By investigating the research questions outlined above, our study contributes to IS research in four ways. First, we advance our understanding of digital nudging by empirically investigating the effectiveness of a novel digital salience nudge in a highly relevant application domain. Our application combines prior IS research on salience and nudging, informs the design of future digital nudging approaches, and highlights potential differences between digital and analogue nudging approaches. Second, we add to salience theory in decision-making by combining visual salience theory from cognitive psychology with attribute salience as defined in behavioural economic theory. Third, we enter the current discussion about the ethical and human dignity-preserving use and potential "dark sides" of information technology by investigating the impact of two measures that have been proposed by prior research to overcome key objections associated with digital nudges. Fourth, our results provide guidance for practitioners planning to design and implement digital nudges in their online sales channels or other digital technologies.

2. Theoretical background

Neoclassical economic theory assumes that humans are rational agents with consistent, stable preferences who scrutinise all the information provided on available alternatives to choose the option providing the highest expected utility (Becker, 1976; Von Neumann & Morgenstern, 1944). By contrast, research in psychology and behavioural economics states that individual preferences are context-sensitive, attention-based, and constructed at the moment of choice (Bordalo et al., 2013; Evans, 2008; Gigerenzer, 2008; Kahneman, 2003). In addition, humans often lack the cognitive resources to process all the available information and instead employ heuristic decision strategies (i.e., mental rules of thumb; Evans, 2008; Gigerenzer, 2008; Kahneman, 2003; Tversky & Kahneman, 1974). Consequently, in situations involving uncertainty, individuals' choices and behaviours are amenable to the influence of the physical, social, and psychological context in which they make decisions (Johnson et al., 2012; Mertens et al., 2022; Thaler & Sunstein, 2008; Thaler et al., 2013).

2.1. Choice architecture and nudging

This susceptibility to seemingly irrelevant aspects of a choice situation can lead to biased decisions and estimates that are at odds with the decision-maker's best interests (Evans, 1989; Kahneman, 2003; Kahneman et al., 1982; Milkman et al., 2009). However, this susceptibility also opens the door for interventions intended to promote individually and socially desirable behaviours by adjusting the decision situation (Thaler & Sunstein, 2008; Thaler et al., 2013). The term *choice architecture* describes the effects of such changes in a decision's structure on behaviour (Johnson et al., 2012; Mrkva et al., 2021; Thaler & Sunstein, 2008). Changes of choice architecture intended to steer people's behaviour are commonly called nudges. A common example of a nudge is the implementation of a pre-selected choice option (i.e., a default); pre-selecting a certain decision alternative can greatly increase its likelihood of being selected (Jachimowicz et al., 2019; Johnson & Goldstein, 2003; Smith et al., 2013; Thaler et al., 2013).

One important general implication of the literature on choice architecture is that there is no neutral way to frame a decision (Johnson et al., 2012; Thaler & Sunstein, 2003, 2008). To put it more bluntly, it is impossible *not* to influence or nudge people with how choices are presented. Therefore, it is inevitable for choice architects (i.e., the agents responsible for organising the decision context; Thaler et al., 2013) to affect people's choices with their decision about the choice environment. Given this ubiquity of choice architecture, the concept of nudging has attracted substantial attention from scholars, organisations, marketers, and policymakers alike (Benartzi et al., 2017; Beshears & Kosowsky, 2020). Nudges thereby often provide an easy-to-implement and inexpensive alternative to conventional interventions such as

strong economic incentives, fines, or bans. In addition to defaults, various other nudges have been shown to induce sizeable changes in human behaviour across different application domains (see Beshears & Kosowsky, 2020; Hummel & Maedche, 2019; Johnson et al., 2012; Mertens et al., 2022, for reviews).

2.1.1. Digital nudging

In the past two decades, people's everyday lives have become increasingly embedded in digital and online contexts, and the fundamental insights of choice architecture and nudging literature in behavioural economics have been applied and extended to digital choice environments. Information systems research has shown that heuristics and biases do not only drive behaviour in analogue contexts but also shape decisions in the digital world (see Arnott & Gao, 2022, for a review). These insights have led IS scholars to adopt the concept of nudging and prompted the emergence of the digital nudging research stream (Mirsch et al., 2017; C. Schneider et al., 2018; Weinmann et al., 2016).

However, online and digital choice environments often differ substantially from analogue decision contexts (e.g., Brynjolfsson et al., 2011; He et al., 2019), leading to important differences in individual behaviour, especially in the consumer domain (see e.g., Brynjolfsson et al., 2011; Darley et al., 2010). First of all, it is unclear whether the mechanisms underlying choice architecture and nudging are readily transferrable to digital choice environments (Lembcke et al., 2019; Meske & Amojo, 2020b; C. Schneider et al., 2018). For example, consumers could be susceptible to different changes in the choice architecture and may have developed other habitual behaviour and decision heuristics (Benartzi & Lehrer, 2015; Dennis et al., 2020; Mirsch et al., 2017; C. Schneider et al., 2018). More importantly, online choice environments are commonly not restricted in terms of size or time, making them highly dynamic and customisable (Adjerid et al., 2018). This offers multiple unique opportunities to implement novel nudging interventions, potentially increasing the effectiveness of analogue ones (Caraban et al., 2019; Dennis et al., 2020; Mirsch et al., 2017; Schmidt & Engelen, 2020; C. Schneider et al., 2018; Weinmann et al., 2016). First, by leveraging user interface (UI) design elements, these environments enable very precise, nuanced, and real-time adjustments of the choice architecture, allowing for the delivery of nudges in an optimised way right at the moment of the decision (C. Schneider et al., 2018). This UI-based approach also makes digital nudges easier, faster, and cheaper to implement than most analogue nudges (Mirsch et al., 2017; C. Schneider et al., 2018; Weinmann et al., 2016). Second, digital choice environments allow for more personalised nudging approaches based on user tracking and more advanced behavioural analyses, potentially boosting their effects (Mills, 2022; Peer et al., 2020). Third, digital nudges can utilise a wide variety of interventions that would not be possible in the analogue world due to restrictions in time and space. Examples include targeted in-app push notifications (Valle et al., 2020) or combining real-time feedback and negative reinforcement (Okeke et al., 2018).

To establish a deeper understanding of choice architecture and nudging in digital choice environments, a considerable body of research has investigated digital nudges in various application domains with promising results (see Acquisti et al., 2017; Bergram et al., 2022; Hummel & Maedche, 2019; Mirsch et al., 2017, for reviews). For privacy and security decisions, for example, digital nudges have been shown to effectively steer people towards choices that align more closely with their long-term preferences (Acquisti et al., 2017; Adjerid et al., 2018; Franz & Benlian, 2022). Positive effects have also been found in other domains such as social media (Bergram et al., 2022) and the organisational context (Haki et al., 2022; D. Schneider et al., 2020). However, a large proportion of the previous studies focused on implementing well-established nudges in digital environments or specifically targeted behaviour that is primarily relevant in these contexts (Bergram et al., 2022; Hummel & Maedche, 2019). While it is important to establish whether conventional nudges also work in digital contexts, the lack of development and evaluation of highly targeted digital nudges that leverage the specific characteristics of these contexts (e.g., dynamic and highly customisable user interfaces) is a missed opportunity. One area that could particularly benefit from such digital nudging approaches is the food choice domain. Here, due to systematic cognitive biases (as outlined above), many consumers struggle to follow through with their intentions and could therefore particularly benefit from the support of digital nudges (Schulte-Mecklenbeck et al., 2013).

2.1.2. Digital nudging to promote healthy food choices

As consumers are increasingly shifting to online sales channels to purchase groceries, the importance of digital choice environments in the food choice domain has grown considerably in the recent past (Verschueren et al., 2021). Consequently, multiple digital nudging approaches to promote healthy food choices have been investigated in IS and other disciplines. For example, studies in the IS domain have focused on default nudges or social norms and showed their general effectiveness in changing online food and recipe choices (e.g., Coffino et al., 2020; Elsweiler et al., 2017; Ingendahl et al., 2021; Jesse et al., 2021; Starke et al., 2021). In addition, a large body of evidence has

revealed that implementing food labels in online supermarkets can lead to improvements in diet quality (Stuber et al., 2022).

In the domain of nutrition research, Shin et al. (2020) describe a notable digital extension of these conventional labelling approaches (i.e., simply displaying food labels on or next to products). Using an experimental online grocery store, the authors allowed consumers to dynamically choose their preferred food label and provided real-time feedback on their shopping basket's current healthiness as well as a healthiness summary of their final grocery cart that included product-level information. These interventions significantly improved the nutritional quality of the food products purchased compared with the control group. However, the authors did not compare their dynamic display and feedback approach with a traditional labelling condition. Another study in this domain showed that front-of-package labels providing information on both the health and environmental impact of food products had the largest effect when the accessibility of relatively healthy and sustainable products was simultaneously increased using a recommendation agent (De Bauw et al., 2022).

These results suggest that digital nudges can be valuable extensions of traditional nudging interventions and that implementing them is a promising avenue to guide consumer decisions towards healthier food products. Because digital choice environments allow for detailed and precise manipulations of the user interface that would not be possible in analogue contexts, a particularly promising approach in this domain is the use of digital salience nudges.

2.2. Digital salience nudging

The theoretical foundation of salience nudges is attentional processes that critically shape human decision-making. Attention defines which parts of people's environments are encoded, processed in conscious thought, and thus integrated into decision processes (Kahneman, 1973). Because human cognitive resources are limited and attention requires cognitive effort, the distribution of attention must be highly selective (Kahneman, 1973). Ideally, attention should only be drawn by goal-directed stimuli that are of high relevance for a decision. However, especially in situations driven by heuristic decision-making, attention is often misdirected towards stimuli that are particularly salient. This can lead to distortions in people's choice behaviour, which is commonly referred to as salience bias (Bordalo et al., 2022; Kahneman et al., 1982; Tiefenbeck et al., 2018). Salience can affect decisions on multiple dimensions, with the effects of visual salience being the most intuitive mechanism.

2.2.1. Visual salience

In digital shopping environments, preattentive, unconscious processes often direct attention towards the most salient visual stimuli (e.g., distinct colour, shape, orientation, or size of a displayed product; Still & Still, 2019). Consequently, these stimuli often receive greater weight in the moment of choice than less salient ones do (Bordalo et al., 2022; Milosavljevic et al., 2012; Mormann & Russo, 2021; Shimojo et al., 2003; Taylor & Thompson, 1982).

In information systems research, the principle of visual salience has been investigated and leveraged in multiple application domains. A great proportion of studies have investigated advertisements or banners and found that their salient visual features such as animations successfully attract users' attention (Ahn et al., 2018; Cheung et al., 2017; Hong et al., 2004, 2021; Sun et al., 2013). Other research has showed that highlighting specific information in online environments attracts visual attention and can affect consumer perceptions and behaviours (Jaeger & Eckhardt, 2021; Kim & Dennis, 2019; Y. Lee & Chen, 2019; Wan et al., 2017; Willems et al., 2019; Yi et al., 2019). These results illustrate and emphasise that the effects of visual salience are of great relevance in digital choice contexts. They further show that salience manipulations are a viable approach to guide attention and thus behaviour.

This amenability of human visual attention to external influences is a promising starting point for the development of interventions that adjust the visual salience of choice alternatives or information to steer individuals towards healthier food choices (Blumenthal-Barby & Burroughs, 2012; Noggle, 2018). The potential of digital salience nudges is further highlighted by the fact that food choice is a highly frequent and habitual behaviour that is largely governed by heuristic processes (Schulte-Mecklenbeck et al., 2013; Torma et al., 2018), making it particularly receptive to salience manipulations (Ares et al., 2020; Wansink, 2006). Prior research in analogue choice contexts has adopted this approach and investigated the effects of salience nudges such as visibility enhancements or food labels (see Cadario & Chandon, 2020; Wilson et al., 2016, for overviews). The results of these studies were mixed and suggest rather small effects of these nudges (Cadario & Chandon, 2020; Wilson et al., 2016). However, this could be due to a suboptimal timing of these interventions (i.e., not sufficiently visible at the exact moment of choice; van der Laan & Orcholska, 2022) and due to the fact that in conventional supermarkets, too many other stimuli distract people's visual attention. Extending these analogue approaches to digital choice environments is therefore a promising avenue for research.

So far, however, research on digital salience nudging in general and in the food choice domain in particular is scarce. In addition, most existing research has reported specific nudging applications related to salience and their effects but did not focus on unravelling the theoretical foundations of their underlying mechanism. In the nutrition domain, for example, Hoenink et al. (2020) implemented coloured frames and arrows to guide consumers towards healthier food products. This nudge alone did not lead to increases in the percentage of healthy purchases. As the authors do not provide any information on the theoretical underpinning or development of their salience nudge, it is conceivable that it was simply not strong enough to affect food choices independently. A comparably broad research stream exists in the domain of online recipe choices supported by artificial intelligence (Elsweiler et al., 2017; Jesse et al., 2021; Starke et al., 2021). Elsweiler et al. (2017) were able to show that specific features (e.g., colourfulness, brightness, or contrast) that determine the attractiveness of a recipe's associated meal pictures are among the most influential factors affecting consumers' choices. Starke et al. (2021) extended these results to a recipe search task and found these visual attractiveness features of the recipes' associated images to be important predictors of consumers' recipe choices. Furthermore, they found that accompanying healthy recipes with pictures rated high on these visual attractiveness features can increase the healthiness (i.e., FSA score) of people's recipe choices. Similiarly, Jesse et al. (2021) increased the visual salience of certain recipes by implementing a coloured background behind the nudged option. They did not find an effect of this intervention on the individuals' decisions. However, the authors do not provide detailed information about their salience nudge or its theoretical underpinning and implemented an incomplete experimental design, which may have affected the internal validity of their findings. While these studies provide valuable empirical evidence from consumer choices in the food domain, it is unclear to what extent the results of studies investigating recipe choices can be transferred to food choices in online supermarkets. Recipes do not include specific product choices and still leave the individual the freedom to deviate from the instructions in many ways, which could significantly impact the preference for one recipe over another.

Berger et al. (2020) conducted a study in the IS domain that analysed the effects of digital nudges on product-level food choices in online supermarkets. They examined how different nudges affect the sustainability of consumers' online food choices. They found inconclusive results for the effects of their graphic labels and statements and even a negative effect of their instantiation of a salience nudge. As the salience manipulations in the label and statement conditions were comparably weak, it is conceivable that they did not leverage the full potential of salience nudges.

With our study, we aim to extend the existing findings and add to the literature on digital salience nudging. We do so by systematically investigating the effects of a simple but promising digital salience nudge that builds on the very fundamental principles of visual salience. The nudge's underlying principle suggests that reducing the brightness or colour intensity of food products' images can effectively decrease their selection by participants (Dai et al., 2020; Milosavljevic et al., 2012). A recent study in cognition research showed in a series of experiments that reducing the visual salience of unhealthy food products by reducing their brightness led to fewer participants choosing these products, independent of their health and taste preferences (Dai et al., 2020). These results indicate a broad effectiveness of this approach and its suitability as a digital nudge. So far, however, this nudge has only been employed in artificial binary food choice tasks. In our design, we extended this research by employing a digital salience nudge in a more realistic food product choice context.

Based on the literature outlined above, we propose that decreasing the brightness of unhealthy food products reduces their visual salience. This should in turn lead to reduced attention directed towards these products and subsequently decrease their weight in the decision-making process (Dai et al., 2020; Itti & Koch, 2001; Milosavljevic et al., 2012; Mormann & Russo, 2021; Shimojo et al., 2003). Conversely, all other food products should receive greater attention from the decision-makers, increasing their weight in the decision-making process (Dai et al., 2020; Itti & Koch, 2001; Milosavljevic et al., 2012; Mormann & Russo, 2021; Shimojo et al., 2003). Consequently, the proportion of unhealthy food products that individuals choose should decline. Following this argument, we hypothesise that

H_1 Reducing the visual salience of unhealthy food products leads to individuals choosing fewer unhealthy food products.

Our digital salience nudge can directly and precisely manipulate the visual salience of particular food products based on certain characteristics. Although related interventions exist in analogue environments, such as placing food products at consumers' eye level in offline supermarkets, these measures are not as precise and nuanced as our intervention because the consumer's gaze still wanders across the shelf and might be attracted by potentially unhealthy food products with particularly colourful packaging.

2.2.2. Attribute salience theory

In addition to the outlined effects of visual salience, behavioural economists have established the concept of attribute salience (Bordalo et al., 2013, 2022). Their key assumption is that in decision situations, not only do the choice alternatives (e.g., different products) differ in their salience, but so do the attributes of these options themselves. For instance, the smell of freshly baked cinnamon cake might not only increase an individual's craving for that particular product; it might simultaneously (at least momentarily) increase the importance the individual assigns to the attribute of tastiness compared to other attributes such as price or healthiness. As this example illustrates, the salience of attributes is also malleable to external factors and highly context-dependent. As in the case of visual salience, salient attributes receive greater attention from the decision-maker and are overweighted in the decision process, while less salient attributes are underweighted (Bordalo et al., 2012, 2022). In contrast to visual salience effects that impact an individual stimulus' decision weight, attribute salience affects the weighting of a whole dimension in a decision situation, which influences the assessment of all available choice alternatives (Bordalo et al., 2022). Bordalo et al. (2022) propose three different channels that define an attribute's salience, namely contrast (i.e., the attribute stands out when a good is compared to its alternatives); surprise (i.e., the attribute is surprisingly different from a memory-based norm); and, like in the case of the cinnamon cake, prominence (i.e., the attribute is highly available to one's senses or memory).

The effects of attribute salience have also been established in IS research. For example, a prior study has shown that simply manipulating the salience of a game's currency's exchange rate can significantly affect players' perceptions of a virtual item's expensiveness based on the numerosity of its price (Huang et al., 2021). Likewise, salient feedback that is not aligned with developers' goals on crowdsourcing platforms can be detrimental to their performance (H. C. B. Lee et al., 2018). Furthermore, the principle has been shown to affect people's decisions on crowdfunding platforms (Weinmann et al., 2022).

Applied to our research, attribute salience adds another potential operating mechanism to the effects of our digital salience nudge. By reducing the visual salience of unhealthy food products, the importance of the attribute "healthiness" in the decision situation might increase overall. This could be the case because individuals will notice the reduced visual salience of the unhealthy food products and contrast them with the other food products in the choice set, potentially increasing the attention drawn to the healthiness attribute (as suggested by the contrast mechanism outlined above; Bordalo et al., 2022). This should increase the decision weight that the attribute healthiness receives. Participants might therefore not only avoid unhealthy food products because of their reduced visual salience but also actively choose particularly healthy products (Bordalo et al., 2022).

In our study, we distinguish three categories of food products: healthy, medium healthy, and unhealthy. Therefore, not choosing unhealthy food products (as suggested by our first hypothesis) does not imply that consumers choose healthy food products as a substitute. Consumers can also switch to medium-healthy food products, which are closer substitutes for unhealthy products, and have the same visual salience as the healthy product options. However, based on the principle of attribute salience, we hypothesise that

H_2 Reducing the visual salience of unhealthy food products does not only lead to individuals substituting them with medium healthy products (as the closest alternative), but also to individuals choosing more healthy food products.

Following our hypothesised effects, implementing a digital nudge that reduces the salience of unhealthy food products has great potential to significantly influence individual food product decisions and increase their healthiness. Considering the power of such nudges to alter human decision-making, a vast body of research in the nudging literature critically reflects on potential negative issues associated with the implementation of such nudges and numerous eminent scholars have emphasised the necessity to identify possible approaches to overcome them (e.g., Leidner & Tona, 2021; Myers, 2021; Sunstein, 2015).

2.3. Ethical concerns of (digital) nudging

Shortly after Thaler and Sunstein (2008) first introduced the concept of nudging to a broader audience, a vivid academic and political debate regarding its moral justification and ethics emerged (see e.g., Bovens, 2009; Fabian & Pykett, 2022; Hansen & Jespersen, 2013; Schmidt & Engelen, 2020). Because digital nudges build upon the same behavioural economic roots as conventional analogue nudging approaches, they are confronted with the same challenges (Acquisti et al., 2017; Lembcke et al., 2019; Meske & Amojo, 2020a; Renaud & Zimmermann, 2018; C. Schneider et al., 2018; Weinmann et al., 2016).

One of the key critiques of nudging evolves from its underlying behavioural principle. Multiple scholars have raised the claim that nudges mainly work by manipulating people's decisions, restricting individual autonomy and freedom of choice (Hansen & Jespersen, 2013; Reijula & Hertwig, 2022; Schmidt &

Engelen, 2020; Vugts et al., 2020). The underlying conjecture is that most nudging approaches exploit biases in human decision-making to directly shape behaviour without the nudged individuals being aware of it (Bovens, 2009; Hansen & Jespersen, 2013; Schmidt & Engelen, 2020). Therefore, they often cannot detect or contest the influences of nudges. Consequently, the individual's actions and choices do not necessarily reflect the outcomes of their own evaluations and reasoning but rather the tactics of the nudging institution (Hausman & Welch, 2010). Another issue that has been raised relates to the critical requirement formulated by Thaler and Sunstein (2008) that nudges need to promote the welfare of nudged individuals and make them better off as judged by themselves. In fact, it is difficult for the nudging institution to identify whether the nudge follows this principle and aligns with the preferences of the decision-maker (Barton & Grüne-Yanoff, 2015; Grüne-Yanoff & Hertwig, 2016; Hausman & Welch, 2010; Sunstein, 2018). Importantly, these arguments also hold for cases in which the choice architect has the nudged individual's best interests in mind because they are not concerned with a decision's outcome but rather directly address questions of human dignity (Leidner & Tona, 2021; Vugts et al., 2020). To overcome these ethical concerns that are directly related to the working mechanisms at the heart of the nudging principle, several scholars have proposed different measures. They commonly revolve around the way nudges are designed and communicated.

2.3.1. Transparency

One measure proposed to alleviate the criticism of manipulation and restriction of autonomy and increase the ethics of nudging interventions is to increase their transparency (Bruns & Paunov, 2021; Hansen & Jespersen, 2013; Lembcke et al., 2019; Sunstein, 2015). By coupling a nudge with a transparency message disclosing its purpose and underlying mechanisms, one might directly avoid subliminal influence, potential restriction of autonomy, and therefore manipulation (Bruns & Paunov, 2021; Loewenstein et al., 2015; Steffel et al., 2016). However, while recent studies in IS have touched upon transparency in the domain of digital nudges (Lembcke et al., 2019; Meske & Amojo, 2020a), it is still unclear how increased transparency affects a digital nudge's effectiveness. On the one hand, it is conceivable that a digital nudge will forfeit some or all of its behavioural effectiveness if it is transparently communicated to the nudged individuals. As outlined above, nudges commonly work by leveraging people's unconscious, automatic thought and decision-making system (Bovens, 2009; Hansen, 2016; Thaler & Sunstein, 2008). Disclosing transparent information about a digital nudge would lead to people becoming aware of its influence and, thus, involve reflective and conscious thought processes (Hansen, 2016; Steffel et al., 2016). As a consequence, the digital nudge would not operate on a subliminal level anymore, which is often considered one of its key mechanisms and might, therefore, lose some of its behavioural impact. In addition, the increased awareness of the digital nudge's potential influence could lead to people trying to resist or even counteract its influence, which would further threaten its behavioural impact (Bruns & Paunov, 2021; Bruns et al., 2018; Loewenstein et al., 2015; Steffel et al., 2016).

Next to this rather pessimistic account, it is also possible that providing transparent information about a digital nudge could enhance its effects on people's behaviour. In their recent working paper, Bruns and Paunov (2021) conducted a systematic review of empirical studies investigating the effects of different transparency measures on the effectiveness of nudges and found that in most cases, the transparent nudges were at least as effective as their opaque counterparts (e.g., Bruns et al., 2018; Loewenstein et al., 2015; Steffel et al., 2016). Based on multiple non-linear logit models to estimate the effects of different transparency components, they showed that disclosing a nudge's underlying mechanism even has a significant, positive effect on its effectiveness. While these preliminary effects should be interpreted with caution and are mostly based on studies in analogue choice environments using a limited set of nudges (i.e., mostly defaults), they suggest that informing consumers about the underlying mechanisms of a digital nudge could not only help to overcome ethical concerns regarding its use but simultaneously even increase its effectiveness. In fact, this assumption aligns well with nudging and salience theory. Providing transparent information about the digital nudge's presence and desired target behaviour is likely to activate additional reflective thought processes and increases the salience of its ultimate goal (Bordalo et al., 2022; Bruns et al., 2018). If this goal aligns with people's preferences, they might consciously decide to actively follow the behaviour prompted by the digital nudge, which is likely to boost its effects (Bruns & Paunov, 2021; Steffel et al., 2016).

In the present study, we empirically investigate this approach in the domain of digital nudging and add to the current discussion of ethical IS (Myers, 2021). To do so, we investigate how disclosing transparent information about the purpose and underlying mechanisms of our digital salience nudge impacts its effectiveness. Based on prior literature, we hypothesise that

H_3 Transparently communicating the purpose and underlying mechanisms of the digital salience nudge increases its effects on the healthiness of consumers' food choices.

A second measure proposed to overcome ethical concerns associated with nudging aims to ensure that nudges work in the nudged individuals' best interests and align with their long-term preferences. This is a critique that also transparent nudges cannot necessarily help to dispel (Steffel et al., 2016). Therefore, to ethically nudge consumers, it might not be sufficient to increase a nudge's transparency but instead be necessary to give individuals the opportunity to adjust the nudge according to their preferences or decide whether they want it to be instantiated at all. This concept was introduced by Reijula and Hertwig (2022) under the term *self-nudging*.

2.3.2. Self-nudging

Self-nudges refer to "empowering interventions that enable people to design and structure their own decision environments" (Reijula & Hertwig, 2022, p. 1) and are particularly facilitated by digital choice environments and new technology (Kozyreva et al., 2020; Krpan & Urbaník, 2021). These nudges involve a conscious and voluntary decision by decision-makers about whether to implement a nudge (Reijula & Hertwig, 2022; Torma et al., 2018). Thus, consumers can be given a tool to overcome self-control issues, for example, and shape a decision's choice architecture to achieve their long-term goals (Reijula & Hertwig, 2022). The idea of self-nudging is therefore not a completely novel one. In the IS discipline, it is related to research on personalisation (e.g., in recommender systems; see e.g., S. S. Li & Karahanna, 2015; T. Li & Unger, 2012, customisation (Jiang & Benbasat, 2007; Werner et al., 2022), and IS for self-regulation, such as wearables supporting health-related behaviours (Chatterjee et al., 2018; Lehrer et al., 2021).

While personalised nudging approaches have been deemed promising to overcome issues of preference alignment and could greatly benefit from digital choice environments (Mills, 2022; Sunstein, 2013), they also face challenges regarding data availability and privacy concerns (Diederich et al., 2022; Mills, 2022). In addition, building nudges that guide consumers towards certain decisions that align with their prior behaviour does not per se solve concerns raised regarding manipulation and freedom of choice. However, offering a digital nudge as a self-nudge overcomes these issues by transferring the agency to implement the nudge to the decision-maker and could even increase its effectiveness (Grüning et al., 2023; Reijula & Hertwig, 2022; Werner et al., 2022).

Prior research in the wearables domain, for example, suggests that these self-selected devices and their accompanying apps and behaviour change techniques can be very effective in promoting physical activity (Lehrer et al., 2021). To understand behaviour change through wearables, Lehrer et al. (2021) propose a self-leadership perspective (Manz, 1986), suggesting that consumers employ self-leadership strategies using technology and that technology can also deploy such strategies to induce behaviour change. Adapted to our context, the self-nudging approach involves that consumers can decide for themselves whether to implement the digital salience nudge. By deciding in favour of the nudge, they can set a healthy food choice goal themselves and an additional cue to follow this goal, which might also increase its salience (Bordalo et al., 2022; Lehrer et al., 2021). In addition, this active decision for the digital nudge's implementation can serve as a precommitment strategy, adding an additional conscious working mechanism to the digital nudge's effects (G. Bryan et al., 2010; Duckworth et al., 2018; Schwartz et al., 2014). More precisely, the presence of the digital salience nudge could serve as a reminder for the individuals to act in line with their precommited goals (i.e., buying healthy food products). This is likely to boost the effectiveness of the digital salience nudge for these individuals compared to those who did not employ these self-guided strategies (Adjerid et al., 2022; Grüning et al., 2023; Lehrer et al., 2021; Reijula & Hertwig, 2022).

On the other hand, however, the self-nudging approach also allows individuals to deliberately decide *against* the implementation of the digital salience nudge. For example, individuals who do not have a strong goal of eating healthier are unlikely to decide to implement the digital salience nudge. These decision-makers would not benefit from the nudge's effects and might even react negatively to the attempted influence on their food product choices (Brehm & Brehm, 1981; Bruns & Paunov, 2021; Bruns et al., 2018; Loewenstein et al., 2015; Steffel et al., 2016). Recent empirical evidence from an experimental economic study by Diederich et al. (2022), indeed, suggests a hard trade-off between ethicality and effectiveness. The authors found that compared to a preset default value, a self-set default value did not lead to increased contributions in a public goods game due to reduced active contributions. This result suggests that while self-nudges might be effective in changing behaviour on an individual level, their aggregated effects might be reduced by the missing or even negative effects among individuals deciding against the implementation of the nudge.

While Diederich et al. (2022) results are not necessarily representative of the effects of a self-nudge in the consumer domain, they provide initial evidence of its aggregated effects. We therefore hypothesise the following:

H_4 Offering the digital salience nudge as a self-nudge decreases its overall effects on the healthiness of consumers' food choices compared to a preset nudge.

3. Methodology

3.1. Experimental design

To test our hypotheses, we conducted a preregistered randomised controlled trial (RCT) survey-based online experiment with 600 participants. In the experiment, participants were asked to shop for food products in different categories of a mock-up online supermarket. We randomly assigned each participant to one of six experimental groups: a control group (*CG*), a control group with transparent information (CG_T), a group with a preset nudge (*N*), a group with a preset nudge and transparent information (N_T), a self-nudging group (*SN*), and a self-nudging group with transparent information (SN_T). In the self-nudging groups, participants had to decide whether they wanted the nudge to be implemented. An overview of our experimental groups is provided in Table 1 (see Appendix B for detailed instructions).

Our *digital salience nudge* intervention consisted of an opaque white layer added to the representation (i.e., tiles) of unhealthy food products in the product overview of the mock-up online supermarket. An 81% opaque white layer was added to the product's picture, and a 61% opaque white layer was added to its accompanying information to guarantee the readability of all information (see Figure 1). To avoid participants mistakenly interpreting the salience nudge as indicating sold-out products, we informed the preset nudging groups that all food products were available for sale and that the reduced colour intensity signalled that a product was unhealthy. As an attention check, we subsequently asked participants in these groups for the reason why some products in the online shop were shown with reduced colour intensity. Participants who did not answer the question correctly were screened out.

In the *transparent information* groups (CG_T, N_T, SN_T), participants received additional information. For the nudge and self-nudging groups (N_T, SN_T), the additional information consisted of disclosures about the purpose and underlying mechanisms of the nudge. We introduced its purpose with a paragraph describing the adverse health consequences of unhealthy food products and an unbalanced diet. For the nudge's underlying mechanisms, we stated that attentional processes often drive food product decisions and that consumers consequently often do not consider the healthiness of the food products they buy. We further informed participants that reducing the colour intensity of unhealthy food products could help them focus on healthier products by decreasing the attention unhealthy ones receive while leaving individuals the freedom to choose the products they want. To control for the impact of receiving information about the adverse health consequences of unhealthy food products, our research design included a control group with transparent information (CG_T), but no nudge or self-nudge.

In the *self-nudging* conditions (*SN*, SN_T), participants were informed that they could decide whether unhealthy food products should be displayed with reduced colour intensity. Prior to their decision, they saw an example of how the nudge would reduce the colour intensity and, depending on their experimental group, additional transparent information about the

Table 1. Overview of experimental groups.

Transparency	Control group	Preset nudge	Self-nudge
Non-Transparent	*CG*	*N*	*SN*
Transparent	CG_T	N_T	SN_T

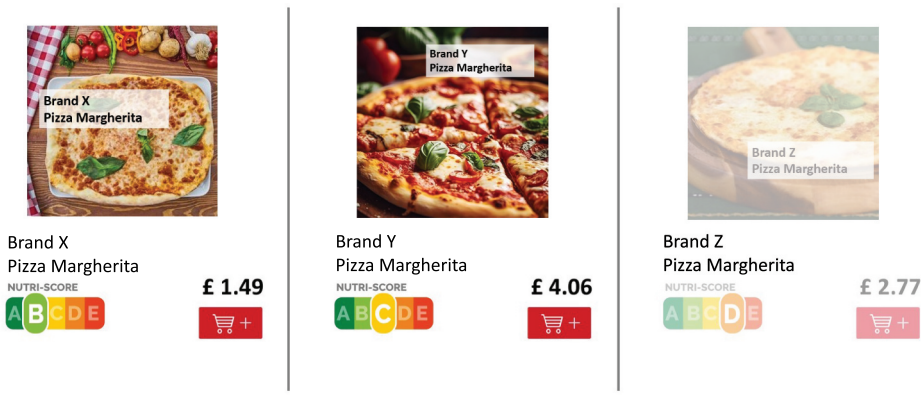

Figure 1. Example tiles of food products classified as healthy (left), medium (in the middle) and unhealthy (right) with the salience nudge implemented.
Note that the food products displayed to the participants contained images from real products from France, Germany, and Spain. For copyright reasons, these product images had to be replaced by royalty-free images without reference to specific brands in the final steps of the article publication process. Exemplary screenshots of the choice sets with the original product images are available on request.

purpose and underlying mechanisms of the nudge. The salience nudge was implemented only if the participants decided in favour of it. In the self- nudging group (*SN*), 61 of 102 participants decided to implement the digital salience nudge (59.8%), while in the self-nudging with transparent information group (SN_T), 70 of 99 (70.7%) participants opted for the implementation.

For the experimental scripts and wording used in the respective conditions, see Appendix B. We implemented an additional attention check in the scales at the beginning of the survey that consisted of a scale item for which participants had to select a specific option (i.e., "It is important that you pay attention to this study. Please tick 'Very much'".). Participants who did not select the correct answer were screened out. The experimental design and procedure of the study received ethical clearance by the German Association for Experimental Economic Research (No. D9ND65xE). In addition, our hypotheses, experimental design, and analyses were preregistered prior to data collection. The preregistration is available under https://osf.io/6ymg2[1]

3.2. Mock-up online supermarket

The mock-up online supermarket was designed and implemented using the survey tool SoSci Survey (https://www.soscisurvey.de/). The online supermarket included six different food product categories (i.e., tomato sauce, chocolate desserts, cream cheese, fruit muesli, biscuits, and frozen pizza margherita) presented to the participants in a random order. In each category, participants were asked to choose one product but could also select an option that they would not purchase any of the products. The product categories were designed to include a variety of both vice and virtue food categories (Hui et al., 2009; Wertenbroch, 1998). Each category comprised nine different food products. The food products were displayed in individual tiles arranged in random order. Each tile (see Figure 1 for an example) consisted of an image of the product's package, the product's name (i.e., brand name and the respective food product category), its Nutri-Score (a five-step coloured nutrition label reflecting a product's nutritional quality as a health indicator; Julia et al., 2015), and its price. The Nutri-Score was included as a healthiness indicator to enable participants to differentiate between the healthiness of the different food products in a category. The Nutri-Score is widely adopted in practice and has been shown to reflect a product's healthiness validly (Julia et al., 2015). Throughout the experiment (i.e., in all instructions displayed to the participants), we referred to products with a Nutri-Score of A or B (i.e., green) as healthy, those with C (i.e., yellow) as medium healthy, and those with D or E (i.e., orange or red) as unhealthy. Participants were therefore aware of this classification and able to identify unhealthy, medium healthy, and healthy food products. This was important to avoid confusion about the products we were referring to when using the terms "healthy", "medium healthy", and "unhealthy".

To avoid the confounding influences of prior beliefs, product preferences, and habit- based decisions, the online supermarket contained only food products with which most participants were not familiar. Because our sample consisted of UK participants, we selected food products from France, Germany, and Spain that were unavailable in the UK. We took advantage of the controlled online environment to control for the influence of prices and imbalances in the product assortment that might otherwise distort consumer decisions. To that end, we selected the products in such a way that in every food product category, a diverse and fully balanced choice set regarding prices and Nutri-Scores was available. Every food product was assigned its true Nutri-Score calculated based on the Open Food Facts database and selected such that in each category, three products had a Nutri-Score of B (light green, healthy), C (yellow, medium healthy), and D (dark orange, unhealthy), respectively. Likewise, the prices assigned to the products were also balanced such that at each Nutri-Score level, one product was available at the low, medium, and high price levels, respectively. The price levels were selected as follows: low (mean price for food products in the respective category at the three largest UK online supermarkets_standard deviation of the price), medium (mean price), and high (mean price + standard deviation). This resulted in a fully balanced 3 (Nutri-Score: green, yellow, orange) x 3 (price: low, medium, high) matrix of products available in every category.

3.3. Participants

We recruited participants using the online participant recruitment service Prolific and restricted our sample to UK residents. We aimed for 100 participants in each of our six randomisation groups and for a total sample size of 600. Seven hundred eighteen individuals started the experiment, 78 (8.4%) of whom were screened out because they answered the attention check regarding the salience nudge incorrectly (see Section 3.1). Additionally, six participants failed the second attention check in the survey, and 34 used unsupported mobile devices and thus had to be excluded from the analysis. Our final sample consisted of 600 participants (see Table 2 for a demographic summary). All the participants who completed the survey received financial compensation.

Our sample comprised a larger proportion of female (63.2%) than male participants (36.8%). This significant imbalance ($\chi^2(1) = 41.15$, $p < .001$) was caused by a female-biased panel from Prolific. To account for this, we included the participants' gender as a control variable in our analyses.

Table 2. Demographic summary of participants.

	CG	CG_T	N	N_T	SN	SN_T
N	99	99	100	101	102	99
Age (*SD*)	36.2 (12.3)	38.6 (13.2)	37.7 (14.1)	35.6 (12.2)	37.2 (12.8)	36.3 (13.2)
NA	-	-	-	1	1	-
Gender Female	64 (64.6%)	65 (65.7%)	60 (60.0%)	51 (50.5%)	67 (65.7%)	71 (71.7%)
Male	35 (35.4%)	34 (34.3%)	40 (40.0%)	50 (49.5%)	34 (33.3%)	28 (28.3%)
NA	-	-	-	-	1	-
Education GSCE	12 (12.1%)	9 (9.1%)	15 (15.0%)	9 (8.9%)	11 (10.8%)	8 (8.1%)
A-Level	19 (19.2%)	17 (17.2%)	19 (19.0%)	20 (19.8%)	18 (17.6%)	15 (15.2%)
Undergraduate	16 (16.2%)	11 (11.1%)	12 (12.0%)	10 (9.9%)	16 (15.7%)	7 (7.1%)
Graduate	37 (37.4%)	47 (47.5%)	35 (35.0%)	40 (39.6%)	33 (32.4%)	49 (49.5%)
Post-graduate	14 (14.1%)	12 (12.1%)	15 (15.0%)	20 (19.8%)	24 (23.5%)	17 (17.2%)
Vocational	1 (1.0%)	3 (3.0%)	4 (4.0%)	2 (2.0%)	0 (0.0%)	3 (3.0%)
JobStudent	13 (13.1%)	6 (6.1%)	15 (15.0%)	13 (12.9%)	11 (10.8%)	8 (8.1%)
Employed	51 (51.5%)	57 (57.6%)	55 (55.0%)	54 (53.5%)	65 (63.7%)	54 (54.5%)
Self-employed	13 (13.1%)	8 (8.1%)	9 (9.0%)	17 (16.8%)	8 (7.8%)	13 (13.1%)
Unemployed	13 (13.1%)	16 (16.2%)	8 (8.0%)	8 (7.9%)	12 (11.8%)	16 (16.2%)
Other	9 (9.1%)	12 (12.1%)	13 (13.0%)	9 (8.9%)	6 (5.9%)	8 (8.1%)

3.4. Measures and scales

The two key dependent variables of our experimental study indicate the share of (un-)healthy food products the participants selected. We operationalised these two variables as follows: We calculated the proportion of unhealthy food products (i.e., products with Nutri-Score D) out of all food products chosen by a participant, where a smaller proportion indicates greater healthiness. Conversely, we also measured the proportion of healthy food products (i.e., products with Nutri-Score B) out of all food products chosen by a participant, with a larger proportion indicating greater healthiness. These two variables are not complementary because participants had the choice between unhealthy, medium healthy, and healthy products or even choosing none of the offered products.

In addition to our two main dependent variables, we included the Brief Self-Control Measure by Tangney et al. (2004), to assess the participants' self control, which has been shown to significantly affect the healthiness of food decisions (de Ridder et al., 2012; Tangney et al., 2004). The scale assesses self-control using 13 items on a 7-point Likert scale, with higher values indicating greater self-control. To assess and control for the participants' current appetite, which is another significant factor influencing food choices (Read & van Leeuwen, 1998), we further included a visual analogue scale (Stubbs et al., 2000) ranging from 0 to 100 (0 = "Not at all hungry"; 100 = "As hungry as I have ever been"). Finally, we also used the single-item Food Choice Questionnaire developed by Onwezen et al. (2019) to assess the strength of the participants' healthy food choice motive on a 7-point Likert scale (1 = "Not at all important"; 7 = "Very important"). All scales used in our experiment are included in Appendix A.

3.5. User study

Prior to participating in the experiment, all participants agreed to the conditions of the study and provided consent to the use of their data. After a short introduction, participants answered demographic questions, questions regarding their food-buying behaviour, and scales regarding their current appetite and self-control (including the second attention check). The next page of the survey described the shopping task. We outlined that participants would be forwarded to an online supermarket in which they should decide on one grocery product in each food category. The introduction further included a cheap talk script asking the participants to behave in the same way they would if they really had to pay for the product they chose and have it delivered (based on Van Loo et al., 2014). We also asked participants to assume that they purchased the food products for themselves and that all packages in each food category contained the same amount of the respective product.

On the next page, an exemplary food product tile was shown, and all parts of the information available for a product were briefly described. Thus, we were able to unobtrusively introduce the Nutri-Score as a healthiness indicator. Participants were further informed that the supermarket featured only a limited assortment of primarily unknown brands and that they could decide not to purchase any of the products. Afterwards, the participants were redirected to the online supermarket and randomly assigned to one of the six experimental groups.

In the online supermarket, the participants received additional information depending on their experimental condition. The control group (*CG*) was welcomed and told that they should decide on one product in each category. In the transparency groups (CG_T, N_T, SN_T), participants were informed about the purpose of the nudge, namely the adverse health consequences of

unhealthy food products (i.e., with a Nutri-Score of D or E) and an unhealthy diet. In the nudge and self-nudging groups with transparent information (N_T, SN_T), participants were also informed that even though they cared about healthy nutrition, many consumers did not pay much attention to the healthiness of food products. The participants then received information indicating that reducing the colour intensity of unhealthy food products could help consumers focus on healthier food products by decreasing the level of attention unhealthy ones received while not restricting the consumer's freedom of choice.

In the preset nudge groups (N, N_T), participants received information that to support its customers, the online supermarket had decided to reduce the colour intensity of unhealthy food products. In the self-nudging groups (SN, SN_T), participants were told that they could decide whether they wanted to see unhealthy food products with reduced colour intensity. In addition, an example image of the reduced colour intensity was shown to all participants in the nudge groups and self-nudging groups (N, N_T, SN, SN_T). Participants in the nudge groups and self-nudging groups were reminded that products shown with reduced colour intensity indicated that these products were rated unhealthy, but were available for purchase. In a subsequent question (i.e., the attention check outlined in Section 3.1), participants in the nudge groups and self- nudging groups were asked to select the reason that some food products were shown with reduced colour intensity. Those Participants who answered this question incorrectly were screened out (8.4%). Afterwards, all participants were shown the different food categories in random order and made their respective food product decisions. In the last part of the survey, they were asked about their shopping experience in the online supermarket.

4. Results

4.1. Randomisation check

To ensure that randomly assigning the participants to the experimental groups led to a balanced distribution of sociodemographic characteristics, we performed multiple randomisation checks. For age (F (5, 592) = 0.70, p = .623), education level ($\chi^2(25) = 25.00$, p = .462), and employment status ($\chi^2(20) = 21.85$, p = .349), no significant differences were revealed. Regarding gender distribution, a chi-squared test revealed a significant difference between the groups ($\chi^2(5) = 11.30$, p = .046). However, a subsequent pairwise test of independence did not show any significant differences ($p > .05$).

Because prior research has shown that a person's current appetite and trait self-control can significantly affect the healthiness of their food product choices, we further calculated one-way between-subject ANOVAs to check whether the groups differed in these variables. There was no difference between the groups in their current appetite (F (5, 594) = 1.41, p = .218). For trait self-control, we observed a significant difference between groups (F (5, 594) = 2.52, p = .048). However, a subsequent post-hoc Tukey test showed that no between-group comparison was significant (all $p > .05$).

4.2. Hypotheses testing

In Table 3, we provide a descriptive overview of the main outcome variables of our experiment. For each dependent variable and hypothesis, we provide detailed inferential analyses in the following sections.

Table 4 summarises the results of our hypotheses tests and gives an overview of the respective regression models that were used to test them.

4.2.1. Proportion of unhealthy food products chosen

We tested our hypotheses regarding the proportion of unhealthy food products chosen by the participants using two linear regression models. Our first model included the following independent variables as predictors of the proportion of unhealthy food products in the participants' shopping baskets: *Nudge* (dummy that equals 1 for the preset nudge groups), *Transparency* (dummy that equals 1 for the transparency groups), *Self-nudge* (dummy that equals 1 for the self-nudging groups), and dummies for the respective interaction terms of the transparency dummy with the nudge and self-nudge dummy variables.[2]

The first model (see Model 1 in Table 5) was significant in predicting the proportion of unhealthy food products (F (5, 594) = 8.73, $p < .001$, $R^2 = 0.06$). It revealed a significant negative main effect of *Nudge*, indicating that participants who experienced the preset nudge reduced the proportion of unhealthy food products in their shopping basket by 12 percentage points (from 19 to 7%) on average compared to the control group. This is equivalent to a reduction of 63% and is in line with our H_1.

Table 3. Descriptive statistics of main outcome variables.

	CG	CG_T	N	N_T	SN	SN_T
Unhealthy 'D' products in % (*SD*)	19.02 (20.48)	12.12 (15.58)	7.17 (14.07)	7.26 (14.42)	12.75 (19.51)	6.57 (12.10)
Healthy 'B' products in % (*SD*)	46.30 (30.54)	52.86 (28.97)	60.33 (27.81)	57.76 (28.14)	53.76 (31.89)	60.44 (30.36)
Total of shopping basket in £ (*SD*)	7.76 (2.60)	7.45 (2.74)	7.66 (2.44)	7.36 (2.16)	7.85 (2.27)	7.21 (2.42)

Table 4. Results of hypotheses tests.

Hypothesis	Regression models	Results
H1	1, 2	Supported
H2	3, 4	Supported
H3	1, 2, 3, 4	Partially supported
H4	1, 2, 3, 4	Partially supported

For *Transparency*, the regression analysis revealed another significant negative main effect. This indicates that informing participants about the adverse health consequences of unhealthy food products and an unbalanced diet in the control group reduced the proportion of unhealthy food products in their shopping baskets by 7 percentage points (from 19 to 12%) compared to the control group without this information. This is equivalent to a reduction of 37%.

The last significant negative main effect of *Self-nudge* indicates that participants in the self-nudging groups reduced the proportion of unhealthy food products in their shopping basket on average by 6 percentage points (from 19 to 13%) on average compared to the control group. This is equivalent to a reduction of 32%.

Model 1 further reveals a significant, positive interaction of (preset) *Nudge* × *Transparency*. This indicates a non-additive effect of the preset digital salience nudge and the transparency intervention. The unstandardised coefficient of this interaction effect is thereby of equal magnitude to the *Transparency* coefficient but indicates an opposite direction. This means that when the preset nudge was implemented, transparently communicating its purpose and underlying mechanisms did not lead to any additional effects. This result is not in line with H_3.

The nonsignificant, very small positive interaction effect of *Self-nudge* × *Transparency*, on the other hand, suggests that for the self-nudging group, providing transparent information about the purpose and underlying mechanisms of the digital salience nudge significantly increased the effect of this intervention on the proportion of unhealthy food products in the participants' shopping baskets. This result is in line with H_3.

To investigate whether the effects of *Nudge* and *Self-nudge* differ, we conducted a Wald test-based comparison of their coefficients using the linearHypothesis function of the car package in R (Fox & Weisberg, 2019; R Core Team, 2023). The test revealed a significant difference (F (1, 594) = 5.91, p = .015), indicating that the effect of the preset nudge on the proportion of unhealthy food products selected by the participants is larger than the effect of offering the digital salience nudge a self-nudge. This result is in line with our H_4.

In addition, we conducted another Wald test-based comparison to investigate whether the linear combination of the coefficients for *Nudge*, *Transparency*, and *Nudge* × *Transparency* differs from the linear combination of the coefficients for *Self-nudge*, *Transparency*, and *Self-nudge* × *Transparency*. We did so to compare the effects of the preset nudge with transparent information and the self-nudge with transparent information. The test did not reveal a significant difference (F (1, 594) = 0.09, p = .763). This suggests that combining the self-nudge with transparent information about its purpose and underlying mechanisms was equally effective in reducing the proportion of unhealthy food products the participants chose to implementing a preset nudge with this transparent information. This result is not in line with our H_4

To test the robustness of these results, we ran a second regression model (Model 2), which included the following control variables: the participant's gender, age, employment status, education level, current appetite, trait self-control, strength of healthy food choice motive, and total shopping basket price.[3]

Model 2 was also significant in predicting the proportion of unhealthy food products (F (20, 576) = 7.74, p < .001, R^2 = 0.18, see Table 5). Like Model 1, it showed three negative significant main effects of our experimental manipulations *Nudge*, *Transparency*, and *Self-nudge*. In Model 2, however, the positive interaction effect of *Nudge* × *Transparency* was only marginally significant (p = .097). We further calculated the same Wald test-based comparisons as for the first model. Again, they revealed a significant difference between the coefficients for *Nudge* and *Self-nudge* (F (576, 1) = 4.64, p = .032) and no significant difference between the linear combinations including

Table 5. Results of regression models for the proportion of unhealthy food products selected by the participants.

	Model 1				Model 2			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.19 (0.02)***	0.00 (0.00)	11.60	<.001	0.26 (0.05)***	0.00 (0.00)	5.18	<.001
Nudge	−0.12 (0.02)***	−0.33 (0.06)	−5.13	<.001	−0.11 (0.02)***	−0.30 (0.06)	−4.92	<.001
Transparency	−0.07 (0.02)**	−0.21 (0.07)	−2.98	.003	−0.05 (0.02)*	−0.16 (0.06)	−2.51	.012
Self-nudge	−0.06 (0.02)**	−0.18 (0.06)	−2.73	.007	−0.06 (0.02)**	−0.17 (0.06)	−2.74	.006
Nudge × Transparency	0.07 (0.03)*	0.16 (0.07)	2.14	.033	0.05 (0.03)$^{+}$	0.11 (0.07)	1.66	.097
Self-nudge × Transparency	0.01 (0.03)	0.02 (0.07)	0.22	.825	0.00 (0.03)	0.00 (0.07)	0.00	.997
Num.Obs.	600				597			
R^2	0.07				0.21			
R^2_{Adj}	0.06				0.18			
F	8.73				7.74			

^{+}p < .1, *p < .05, **p < .01, ***p < .001; B refers to unstandardised and β to standardised coefficients. Note: Model 2 includes control variables.

Transparency and interaction effects (F (576, 1) = 0.04, p = .835), indicating that the self-nudge alone was less effective than the preset nudge, but equally effective if combined with transparent information. In sum, the results of Model 2 corroborate the results of Model 1.

To further scrutinise the robustness of our results, we calculated separate models for the preset nudge (Table C1) groups and for the self-nudge groups (Table C2). The results are reported in Appendix C and support the findings of Model 1 and 2.

4.2.2. Proportion of healthy food products chosen

To investigate our H_2 and establish a more holistic understanding of the effects of our experimental manipulations, we investigated the share not only of unhealthy products but also of healthy products in participants' shopping baskets (once again, note that these two measures are not complementary given the three healthiness levels (healthy, medium healthy, and unhealthy); Section 3.4). We, therefore, calculated a second set of linear regression models (Models 3 and 4). These models predicted the proportion of healthy food products in the participants' shopping baskets, based on the same independent variables as in our first set of models.

Model 3 without control variables (see Table 6) was significant in predicting the proportion of healthy food products (F (5, 594) = 3.32, p = .006, R^2 = 0.02). The model revealed a significant positive main effect of *Nudge* on the share of healthy products. This indicates that implementing our digital salience nudge as a preset nudge increased the proportion of healthy food products by 14 percentage points (from 46 to 60%). This is equivalent to an increase of 30% compared to the control group and in line with our H_2.

In contrast to the models regarding the proportion of unhealthy food products, however, the third model did not reveal a significant main effect of *Transparency*. This suggests that informing participants about the adverse health consequences of unhealthy food products and an unbalanced diet in the control group did not affect the proportion of healthy food products they bought.

For *Self-nudge*, the model revealed a marginally significant main effect, suggesting that on average, participants in the self-nudging groups increased the proportion of healthy food products in their shopping basket by over 7 percentage points compared to the control group (from 46 to 53%). This is equivalent to an increase of 15%.

The model did not reveal any significant interaction effects, suggesting that informing participants about the purpose and underlying mechanisms of the digital salience nudge did not affect its impact on the proportion of healthy food products selected by the participants. This result is not in line with our H_3.

To investigate whether the effects of the preset *Nudge* and *Self-nudge* in Model 3 differ, we conducted a Wald test-based comparison of their coefficients. The test revealed a nonsignificant difference (F (1, 594) = 2.48, p = .116), indicating that both interventions had equal (not significantly different) effects on the proportion of healthy food products the participants selected. This result is not in line with our H_4.

We conducted another Wald test-based comparison to investigate whether the linear combination of the coefficients for *Nudge*, *Transparency*, and *Nudge* × *Transparency* differs from the linear combination of the coefficients for *Self-nudge*, *Transparency*, and *Self-nudge* × *Transparency*. The test did not reveal a significant difference (F (1, 594) = 0.41, p = .523). This suggests that combining the self-nudge with transparent information about its purpose and underlying mechanisms was equally effective in increasing the proportion of healthy food products the participants chose to implementing a preset nudge with this transparent information. This result is also not in line with our H_4.

To test the robustness of these results, we ran a second regression model (Model 4 in Table 6) that included the same control variables as in the regression for the proportion of unhealthy food products. The model was significant in predicting the proportion of unhealthy food products (F (20, 576) = 4.17, p < .001, R^2 = 0.10). It showed a significant main effect of *Nudge* but no other significant effects for our experimental manipulations. The main effect of *Self-*

Table 6. Results of regression models for the proportion of healthy food products selected by the participants.

	Model 3				Model 4			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.46 (0.03)***	0.00 (0.00)	15.53	<.001	0.24 (0.10)*	0.00 (0.00)	2.53	.012
Nudge	0.14 (0.04)***	0.22 (0.07)	3.34	<.001	0.13 (0.04)**	0.20 (0.06)	3.09	.002
Transparency	0.07 (0.04)	0.11 (0.07)	1.56	.120	0.05 (0.04)	0.08 (0.07)	1.21	.228
Self-nudge	0.07 (0.04)$^+$	0.12 (0.07)	1.78	.075	0.07 (0.04)	0.11 (0.06)	1.63	.103
Nudge × Transparency	−0.09 (0.06)	−0.11 (0.07)	−1.54	.124	−0.07 (0.06)	−0.09 (0.07)	−1.23	.218
Self-nudge × Transparency	0.00 (0.06)	0.00 (0.07)	0.02	.985	0.01 (0.06)	0.02 (0.07)	0.23	.818
Num.Obs.	600				597			
R^2	0.03				0.13			
R^2_{Adj}	0.02				0.10			
F	3.32				4.17			

^+p < .1, *p < .05, **p < .01, ***p < .001; B refers to unstandardised and β to standardised coefficients. Note: Model 4 includes control variables.

Nudge was close to being marginally significant, but its p-value remained above the $\alpha = .10$ threshold, indicating that the self-nudge intervention did not lead to a significantly larger proportion of healthy food products in the participants' shopping baskets. However, the result of the Wald test-based comparison of the effect of the preset nudge coefficient with the self-nudge coefficient remained insignificant (F (1, 576) = 2.07, p = .151). In addition, the Wald test-based comparison of the linear combinations of the preset nudge coefficient and the self-nudge coefficient with the transparency coefficient also remained insignificant (F (1, 576) = 0.40, p = .529), supporting the results of Model 3. For the proportion of healthy food products selected by the participants, we also calculated additional separate models for the preset nudge (Table C3) and self-nudge (Table C4) groups to validate the robustness of our results. These are reported in Appendix C and support the results of Model 3 and 4.

5. Discussion

People's everyday lives and decisions increasingly occur in digital environments. In these spheres, consumers are particularly prone to being guided by heuristics and bi- ases in their decision-making processes (Arnott & Gao, 2022; Benartzi & Lehrer, 2015; Dennis et al., 2020; Mirsch et al., 2017), which can lead to choices that are not in their best interests. Based on behavioural economics research and nudging theory, IS scholars have proposed the use of digital nudges to support individuals in overcoming these issues and aligning consumer choices in the digital world with their long-term preferences (Mirsch et al., 2017; C. Schneider et al., 2018; Weinmann et al., 2016). Digital nudges can thereby leverage the almost unlimited customisation possibilities that user interfaces offer, for example, which paves the way for novel and potentially more effective nudging interventions compared to analogue choice environments (Mirsch et al., 2017; C. Schneider et al., 2018).

In this study, we have assessed the potential of digital nudges by proposing and evaluating the effects of a digital salience nudge to guide consumers towards healthier food product choices. Our nudge thereby builds on and combines insights from visual and attribute salience theory obtained in multiple disciplines (Bordalo et al., 2013, 2022; Dai et al., 2020; Itti & Koch, 2001; Milosavljevic et al., 2012; Shimojo et al., 2003; Taylor & Thompson, 1982). We assessed its effects in a preregistered online experiment with 600 participants. The results reveal that our proposed digital salience nudge is effective and leads to a reduction of 63% in the proportion of unhealthy food products selected and to an increase of 30% in the proportion of healthy food products chosen. Combining research on visual and attribute salience as well as on digital nudging therefore seems to be a valuable approach to guide consumer decisions in online shopping contexts.

In addition, we investigated the effect of two concrete measures that have been proposed to overcome ethical concerns about nudges. In doing so, we contribute to the vivid discussion about the ethical use and potential dark sides of information technology (Myers, 2021) and human dignity in digitalisation (Leidner & Tona, 2021). More specifically, we evaluated the impact of increasing the transparency of our nudge and of the self-nudging approach on the effectiveness of our digital salience nudge. According to the results of our experiment, transparently communicating the purpose and underlying mechanisms of the digital salience nudge only increases its effects if combined with a self-nudging intervention, and only for the proportion of unhealthy food products selected.

As we expected, offering the digital salience nudge as a self-nudge decreases its effectiveness compared to a preset nudging approach. However, combining it with transparent information seems to restore the effects to an equal level with the preset nudge. These results have various implications for both theory and practice and contribute to the ongoing ethical discussion regarding the effectiveness and ethics of digital nudges.

5.1. Theoretical contributions and implications

Prior research in psychology, neuroscience, and economics has shown that salience effects are a key driver of human attention, behaviour, and decision-making (Bordalo et al., 2022; Dai et al., 2020; Itti & Koch, 2001; Milosavljevic et al., 2012; Shimojo et al., 2003). Various studies in IS research have transferred these results to digital choice environments (e.g., Hong et al., 2021; Y. Lee & Chen, 2019; Sun et al., 2013). While it is well established that salience manipulations are a viable approach to influence individual behaviour and decisions (Blumenthal-Barby & Burroughs, 2012; Noggle, 2018), digital nudging approaches that leverage salience effects to support consumers in acting in line with their long-term preferences are still scarce. So far, prior research on digital salience nudging has mostly focused on salience as a part of broader nudging approaches, produced inconclusive results, and not elaborated on the underlying the oretical mechanisms of salience (Berger et al., 2020; Franz & Benlian, 2022; Jaeger & Eckhardt, 2021; Jesse et al., 2021). Our study extends this research and contributes to prior literature on digital nudging. We provide evidence that applying the broad knowledge of visual salience effects in digital choice environments is a promising approach to inform the development of digital nudges. By adapting the results by Dai et al. (2020) to the context of digital nudging and drawing

on salience theory from psychology, neuroscience, and economics, we further showed that integrating and combining fundamental principles of other research domains can be fruitful for the development of effective digital nudges.

To date, a large proportion of IS research on salience has focused on the attention-capturing effects of visual salience (e.g., Ahn et al., 2018; Cheung et al., 2017; Hong et al., 2021; Sun et al., 2013; Willems et al., 2019). Importantly, these salience effects work on the level of individual stimuli (e.g., by highlighting particular products in each given choice set) and affect the weight of these stimuli in decision situations. While they are also likely to be the primary working mechanism of our digital salience nudge (Dai et al., 2020; Milosavljevic et al., 2012), we extended our understanding of this approach by incorporating the concept of attribute salience as another potential mode of action (Bordalo et al., 2013, 2022). Our findings suggest that our digital salience nudge did not only affect the choice of unhealthy products, which were displayed in reduced salience. Rather than merely switching from unhealthy products to medium healthy products as their closest substitutes, participants considerably increased the share of healthy products in their shopping baskets. These findings indicate that in addition to affecting the salience at the product level, our nudge increased the salience of the attribute healthiness for the food choice situation. Thus, our findings complement the extant research on digital salience nudging (e.g., Berger et al., 2020; Franz & Benlian, 2022; Jesse et al., 2021; Kim & Dennis, 2019) that so far has largely overlooked attribute salience effects. We propose that visual and attribute salience effects go hand in hand, which suggests that visual salience manipulations, which are easy to implement in digital choice environments, can be instrumentalised to also increase the salience of specific attributes in a decision situation (Bordalo et al., 2013, 2022; Kim & Dennis, 2019). Replicating these findings and verifying our proposed theoretical explanations for them is a promising endeavour for future research.

In analogue choice environments, the results of prior research indicate that salience-based nudges such as food labels and visibility enhancements lead to comparably weak effects on food choice behaviour (Cadario & Chandon, 2020; Wilson et al., 2016). The results of our study suggest that digital salience nudges could be a promising extension of their analogue counterparts. We implemented our digital salience nudge on top of a well-known food label (i.e., the Nutri-Score) that was present in all conditions. Thereby, the food label itself should already promote healthier food choices as suggested by prior research (Shin et al., 2020; Stuber et al., 2022) and thus raise the bar for the comparative effectiveness of our digital salience nudge. Compared to the control group, in which participants were exposed to the food label only, however, enhancing the food label with the digital salience nudge was significantly more effective. This finding indicates that the precise, targeted adjustments of the information flow towards the end users enabled in digital choice environments (Dennis et al., 2020; Mirsch et al., 2017; C. Schneider et al., 2018) could enhance the effects of conventional nudging approaches. While Hummel and Maedche (2019) have found that digital and analogue nudges are generally equally effective, prior research has also suggested that there might be differences between analogue and digital choice environments in the effectiveness of nudges (Dennis et al., 2020; C. Schneider et al., 2018). Of course, our findings are associated with certain caveats and should be interpreted accordingly (see Section 5.3). Nevertheless, the results of our study encourage future research to investigate these potentially differential effects and their underlying causes in more detail.

We further contribute to the ongoing debate in the digital nudging domain that has questioned the ethical underpinnings of digital nudging interventions and discussed potential measures to overcome them (Gray et al., 2018; Lembcke et al., 2019; Meske & Amojo, 2020a; Renaud & Zimmermann, 2018; Weinmann et al., 2016). More specifically, we investigated how disclosing transparent information about the purpose and underlying mechanisms of our digital salience nudge impacts its effects. Our results show that a higher level of transparency significantly increased the nudge's effectiveness only in one specific case (for the proportion of unhealthy food products in the self-nudging group), which is not in line with Bruns and Paunov's (2021) results. In fact, in all other cases, our transparency intervention did not affect the digital salience nudge's effectiveness at all (i.e., neither positively nor negatively), which is in line with the results of prior research in other contexts (Bruns et al., 2018; Loewenstein et al., 2015; Steffel et al., 2016). Our findings therefore indicate that designing digital nudges in more ethical ways (Lembcke et al., 2019; Meske & Amojo, 2020a) does not come at the cost of reducing their impact. This result is important for digital nudging theory and the development of ethical IS interventions as well as respecting human dignity in a digitalised world (Leidner & Tona, 2021; Myers, 2021). In summary, our findings suggest that disclosing transparent information about a digital nudge does not decrease its effectiveness and is thus a viable approach to overcoming ethical concerns (primarily raised in the academic community rather than by practitioners).

Finally, we investigated how offering our digital salience nudge as a self-nudge instead of a preset nudge affects its effectiveness. The concept of self-nudging is recent (Reijula & Hertwig, 2022; Torma

et al., 2018), and self-nudging is facilitated by the flexibility of digital choice environments (Kozyreva et al., 2020; Krpan & Urbaník, 2021; Reijula & Hertwig, 2022). We conducted one of the first empirical studies on self-nudging in the consumer domain. Based on our empirical study, compared to a preset implementation, the effectiveness of our digital salience nudging intervention when offered as a self-nudge is reduced in its overall impact on unhealthy food product choices. This finding supports the results of Diederich et al. (2022). However, combining it with transparent information about its purpose and underlying mechanisms as suggested by Reijula and Hertwig (2022), leads to effects similar to those of a preset implementation. These results indicate that combining transparency and self-nudging is a promising approach to overcoming the ethical objections and concerns associated with digital nudges (that at least exist in the academic community) without compromising the effectiveness of the approach. The fact that between 30 and 40% of participants in the self-nudging groups decided against the implementation of the digital salience nudge further emphasises that preset nudging interventions are not always in consumers' best interests (Barton & Grüne-Yanoff, 2015; Gru¨ne-Yanoff & Hertwig, 2016; Hausman & Welch, 2010; Sunstein, 2018). In contrast, self-nudges take heterogeneous individual preferences into account and are a promising extension of the digital choice architect's toolbox. We therefore encourage future research to investigate additional application scenarios for self-nudging.

5.2. Practical implications

Beyond these contributions to academic research, our results have multiple implications for practitioners. First, we have provided evidence that digital salience nudges are a valuable approach to guiding consumers in online decision-making contexts and ecommerce environments in particular. However, it is important that practitioners always respect the fundamental principle of nudging, which states that nudges should always be implemented in such a way that they increase the self-perceived welfare of the nudged individuals.

Second, we found that disclosing information about the purpose and underlying mechanisms of a nudge does not alter its effectiveness. This suggests that practitioners can be open to communicating information about the digital salience nudges implemented in their product offerings and might thus be able to directly address potential ethical objections held by third parties without impairing the effectiveness of the nudge.

Third, we introduced the concept of self-nudging as a promising approach to directly integrate consumers into the implementation of behavioural interventions in online choice environments. Combined with disclosing transparent information about the nudge, self-nudging opportunities are a promising approach to supporting consumers in making online decisions in line with their long-term preferences. This could also lead to beneficial effects for the agents offering these self-nudges, as recent research in web customisation suggests (Werner et al., 2022).

However, one must consider that our specific application of a digital salience nudge to support consumers in making healthier food choices might lead to irritation and resistance among food suppliers if introduced in online supermarkets. An intervention discouraging people from buying certain (in this case, unhealthy) products is not necessarily in the interest of all suppliers, including if the intervention is offered as a self-nudge. Therefore, our approach could also be adopted by third-party organisations developing client-side applications to adjust the interface of ecommerce environments to support consumers. More specifically, Mayer et al. (2021) recently introduced the concept of guerilla modding that could serve as a starting point for such endeavours.

On a more abstract level, the results of our study could also be adapted to other application domains and electronic platforms. Digital salience nudges can be used to guide individuals on news platforms or social media by adjusting the visual salience of sensitive or less relevant content, for example (Kim & Dennis, 2019). The distinction between reduced-salience content and content shown in a regular format could, for example, be based on the same mechanisms and data that currently drive web personalisation efforts (Ho & Lim, 2018; Tam & Ho, 2005). Building on digital salience nudges, these platforms would not censor or block content but could still provide users with a soft prefilter option that can easily be overruled by directing attention to the items reduced in visual salience. Another possible application domain is e-finance platforms that could, for example, use digital salience nudges to guide customers towards sustainable investment opportunities.

5.3. Limitations and future research

Our study provides valuable insights into digital salience nudges and guidance for the ongoing ethical discussion of the potentially patronising nature of digital nudging mechanisms. However, we acknowledge several limitations of our study that could open avenues for future research.

First, we tested our hypotheses based on an online experiment that included a fictitious food choice task in a mock-up online supermarket with whose product assortment the participants were unfamiliar. The participants' choices therefore had no real consequences

because the products were not delivered to them and they did not have to spend real money. While this is a common experimental paradigm to obtain first estimates of effect sizes, future studies must investigate our nudge's effects in more realistic, incentive-compatible experiments such as field studies. It is conceivable that the effects of the digital salience nudge would be smaller in these designs. For the unknown products in our study, the participants were likely to rely solely on the information provided to them (because they did not have any prior experiences to draw from). In shopping contexts that contain brands and products consumers are familiar with, habits and brand perceptions will have a greater influence on decisions, which might limit the nudge's effects.

Second, the results regarding the self-nudging intervention should be interpreted with caution. As emphasised above, participants selected whether they wanted the nudge to be implemented. It is likely that only people whose preferences or goals align with the self-nudge's intended effects will implement it. While this is one of the key advantages of a self-nudge compared to a preset nudge, the self-nudge by nature also gives rise to potential self-selection effects that might bias the intervention's impact. We carefully accounted for these effects by analysing the effects on a treatment group basis including participants who decided against the implementation and controlling for participants' healthy eating goals in our models. Furthermore, in Appendix D, we report the results of instrumental variable regression models that accounted for the endogenous treatment effects in the self-nudging groups. The results of these models fully support our effects.

Third, prior survey research has shown that consumers do not necessarily share the critique that scholars have raised regarding the ethical underpinnings of nudging. In fact, most consumers do not seem to object to the implementation of nudges, as recent evidence has shown (Michaelsen et al., 2021; Sunstein, 2016; Sunstein et al., 2019). While most of these results were obtained using vignette scenarios and did not focus on digital nudging in particular, they suggest that there might be a gap between academic discussions about ethics and actual consumer perceptions and preferences. Our proposed measures to overcome ethical objections might therefore not have practical relevance for consumers' reactions to nudges but rather address scholarly critique. To further illuminate this potential gap, future research could further investigate consumer reactions to nudges apart from first-order behavioural effects.

Fourth, our study focused only on one specific application of our proposed digital salience nudge. While promoting healthier food choices is one of the most common application scenarios in nudging research, our results should be validated in other contexts and choice domains. In addition, it would be beneficial to investigate how self-nudging and personalised nudging approaches (e.g Mills, 2022) could be combined to provide consumers with an optimised and customised nudging experience.

Finally, we answered our research questions and derived our hypotheses by combining general nudging and choice architecture theory with different aspects of salience theory. While this approach was a viable starting point to guide our research, it also revealed that we are currently missing a holistic, overarching framework to evaluate and structure the effects of digital nudges. Combining multiple aspects of digital nudging, such as necessary preconditions and moderators of a digital nudge's behavioural impact as well as the collateral consequences associated with its use into a more comprehensive theoretical account could therefore be a particularly valuable avenue for future research. Approaches in this realm are likely to benefit from current discussions in behavioural economics and nudging research that have asked for a more thorough investigation of the boundary conditions of nudging interventions (C. J. Bryan et al., 2021; DellaVigna & Linos, 2022; Szaszi et al., 2018, 2022).

Notes

1. We initially hypothesised that the effects of our digital salience nudge on the proportion of (un-)healthy food products would be greater for individuals who implemented it as a self-nudge. We changed this hypothesis for a more holistic view, but still report the results of the instrumental variable regressions testing these hypotheses in Appendix D. We do not report our results on the effects of our interventions on the participants' acceptance of the nudge in this paper.
2. The implementation of the nudge in the self-nudging groups was endogenous. To account for self-selection effects, we additionally calculated instrumental variable regressions that fully supported our main findings and are reported in Appendix D.
3. We did not include the prices of the individual products in our model because the food products in all categories were balanced across three price and healthiness levels and we only analysed the food product choices on an aggregated level.

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Appendix A. Scales

A.1. *Brief Self-Control Measure*

Please indicate how much each of the following statements reflects how you typically are.

(1) I am good at resisting temptation.
(2) I have a hard time breaking bad habits.
(3) I am lazy.
(4) I say inappropriate things.
(5) I do certain things that are bad for me, if they are fun.
(6) I refuse things that are bad for me.
(7) I wish I had more self-discipline.
(8) People would say that I have iron self-discipline.
(9) Pleasure and fun sometimes keep me from getting work done.
(10) I have trouble concentrating.
(11) I am able to work effectively towards long-term goals.
(12) Sometimes I can't stop myself from doing something, even if I know it is wrong.
(13) I often act without thinking through all the alter

Answer format: 7-point Likert scale (1 = "Not at all"; 7 = "Very much")

A.2. *Current Appetite*

Please indicate with the slider the position on the line which best corresponds to your feelings at the moment.

(1) How hungry do you feel?

Answer format: Sliding tool (0 = "Not at all hungry"; 100 = "As hungry as I have ever been")

A.3. *Healthy food choice motive*

It is important to me that the food I eat on a typical day. . .

(1) is healthy.

Answer format: 7-point Likert scale (1 = "Not at all important"; 7 = "Very important")

Appendix B. Experimental scripts first introduction page for all groups

For the following task, we will forward you to an online supermarket. There, we would like you to shop for grocery products in the following food categories:

- Fruit muesli
- Chocolate desserts
- Biscuits
- Cream cheese
- Frozen pizza margherita
- Tomato sauce

You will be asked to decide on one product in each category.
We would like you to behave in the same way that you would if you really had to pay for the product you choose and get it delivered. Please take into account how much you would really want the product, as opposed to other alternatives. Please act for each of the following decisions as if you were in a real online supermarket.
Please assume that you are buying the food products for yourself and that all packages in each food category contain the same amount of the respective product (e.g. all muesli packages contain 750 g of muesli).

B.1. *Second introduction page for all groups*

The online supermarket features only a limited assortment with primarily unknown brands. The products will be shown in the format pictured below.
To select a product, simply click on it and confirm your choice at the bottom of the page. There, you will also find an option if you do not want to buy any of the products.

B.2. *Task introduction control group*

Welcome to our online supermarket!
Thank you very much for choosing us for your grocery shopping today.
On the next pages, we will ask you to choose among a set of products in our store. For each product category, you are supposed to choose one item.

B.3. *Task introduction control group with transparent information*

Welcome to our online supermarket!
Thank you very much for choosing us for your grocery shopping today.
We care about our customers. Therefore, we would like to inform you about recent scientific research that has shown that unhealthy food products (i.e., with a Nutri-Score of D or E) and an unbalanced diet are among the main causes of non-communicable diseases (e.g., diabetes, heart diseases etc.). These diseases account for over 20% of global deaths.
On the next pages, we will ask you to choose among a set of products in our store. For each product category, you are supposed to choose one item.

B.4. *Task introduction preset nudge group*

Welcome to our online supermarket!
Thank you very much for choosing us for your grocery shopping today.
To support our customers, we decided to reduce the colour intensity of all unhealthy food products (i.e., with a Nutri-Score of D or E). See the picture below for an example. On the next pages, we will ask you to choose among a set of products in our store. For each product category, you are supposed to choose one item.
Please remember that if a product is shown with reduced colour intensity, it is considered unhealthy but is still available for purchase.

B.5. *Task introduction preset nudge group with transparent information*

Welcome to our online supermarket!
Thank you very much for choosing us for your grocery shopping today.
We care about our customers. Therefore, we would like to inform you about recent scientific research that has shown that unhealthy food products (i.e., with a Nutri-Score of D or E) and an unbalanced diet are among the main causes of non-communicable diseases (e.g., diabetes, heart diseases etc.). These diseases account for over 20% of global deaths.
Many people say that they are overwhelmed by the number of product features when they buy food products. As a result, even consumers who care about healthy nutrition often do not pay much attention to the healthiness of the food products they buy. Reducing the colour intensity of unhealthy food products might help customers to focus on healthier products by decreasing the amount of attention unhealthy ones receive – yet leaving consumers the freedom to choose the products they want.
To support our customers, we decided to reduce the colour intensity of all unhealthy food products (i.e., with a Nutri-Score of D or E). See the picture below for an example. On the next pages, we will ask you to choose among a set of products in our store. For each product category, you are supposed to choose one item.
Please remember that if a product is shown with reduced colour intensity, it is considered unhealthy but is still available for purchase.

B.6. *Task introduction self-nudging group*

Welcome to our online supermarket!
Thank you very much for choosing us for your grocery shopping today.
On the next page, you can decide if you want to see unhealthy food products in our online supermarket with reduced colour intensity. Here is an example:
On the next pages, we will ask you to choose among a set of products in our store. For each product category, you are supposed to choose one item.
Please remember that if a product is shown with reduced colour intensity, it is considered unhealthy but is still available for purchase.
You can now decide: would you like to reduce the colour intensity of all unhealthy food products in our online supermarket? (Yes/No)

B.7. *Task introduction self-nudging group with transparent information*

Welcome to our online supermarket!
Thank you very much for choosing us for your grocery shopping today.
We care about our customers. Therefore, we would like to inform you about recent scientific research that has shown that unhealthy food products (i.e., with a Nutri-Score of D or E) and an unbalanced diet are among the main causes of non-communicable diseases (e.g., diabetes, heart diseases etc.). These diseases account for over 20% of global deaths.

Many people say that they are overwhelmed by the number of product features when they buy food products. As a result, even consumers who care about healthy nutrition often do not pay much attention to the healthiness of the food products they buy. Reducing the colour intensity of unhealthy food products might help customers to focus on healthier products by decreasing the amount of attention unhealthy ones receive – yet leaving consumers the freedom to choose the products they want.

On the next page, you can decide if you want to see unhealthy food products in our online supermarket with reduced colour intensity. Here is an example:

On the next pages, we will ask you to choose among a set of products in our store. For each product category, you are supposed to choose one item.

Please remember that if a product is shown with reduced colour intensity, it is considered unhealthy but is still available for purchase.

You can now decide: would you like to reduce the colour intensity of all unhealthy food products in our online supermarket? (Yes/No)

Appendix C. Separate regression models for preset nudge and self-nudging groups

Table C1. Results of regression models for the proportion of unhealthy food products selected by the participants for the preset nudging groups.

	Model 5				Model 6			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.19 (0.02)***	0.00 (0.00)	11.59	<.001	0.26 (0.06)***	0.00 (0.00)	4.32	<.001
Nudge	−0.12 (0.02)***	−0.35 (0.07)	−5.12	<.001	−0.10 (0.02)***	−0.30 (0.06)	−4.79	<.001
Transparency	−0.07 (0.02)**	−0.20 (0.07)	−2.97	.003	−0.05 (0.02)*	−0.16 (0.06)	−2.47	.014
Nudge × Transparency	0.07 (0.03)*	0.18 (0.08)	2.14	.033	0.05 (0.03)	0.13 (0.08)	1.71	.088
Num.Obs.	399				398			
$_R2$	0.08				0.25			
R^2Adj	0.07				0.22			
F	11.67				7.19			

$^{+}p < .1$, $p < .05$, $p < .01$, $p < .001$; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 6 includes control variables.

Table C2. Results of regression models for the proportion of unhealthy food products selected by the participants for the self-nudging groups.

	Model 7				Model 8			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.19 (0.02)***	0.00 (0.00)	10.97	<.001	0.27 (0.07)***	0.00 (0.00)	4.17	<.001
Self-nudge	−0.06 (0.02)*	−0.18 (0.07)	−2.58	.010	−0.06 (0.02)*	−0.17 (0.07)	−2.50	.013
Transparency	−0.07 (0.02)**	−0.19 (0.07)	−2.81	.005	−0.05 (0.02)*	−0.15 (0.07)	−2.34	.020
Self-nudge × Transparency	0.01 (0.03)	0.02 (0.08)	0.21	.834	0.00 (0.03)	−0.01 (0.08)	−0.09	.929
Num.Obs.	399				397			
$_R2$	0.06				0.21			
R^2Adj	0.05				0.17			
F	8.63				5.53			

$^{+}p < .1$, $p < .05$, $p < .01$, $p < .001$; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 8 includes control variables.

Table C3. Results of regression models for the proportion of healthy food products selected by the participants for the preset nudging groups.

	Model 9				Model 10			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.46 (0.03)***	0.00 (0.00)	15.95	<.001	0.27 (0.11)*	0.00 (0.00)	2.40	.017
Nudge	0.14 (0.04)***	0.24 (0.07)	3.43	.001	0.13 (0.04)**	0.22 (0.07)	3.27	.001
Transparency	0.07 (0.04)	0.11 (0.07)	1.60	.110	0.06 (0.04)	0.10 (0.07)	1.45	.149
Nudge × Transparency	−0.09 (0.06)	−0.14 (0.09)	−1.58	.115	−0.08 (0.06)	−0.12 (0.08)	−1.41	.158
Num.Obs.	399				398			
$_R2$	0.03				0.14			
R^2Adj	0.03				0.10			
F	4.55				3.34			

Note: $^{+}p < .1$, $p < .05$, $p < .01$, $p < .001$; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 10 includes control variables.

Table C4. Results of regression models for the proportion of healthy food products selected by the participants for the self-nudging groups.

	Model 11				Model 12			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.46 (0.03)***	0.00 (0.00)	15.12	<.001	0.20 (0.12)$^{+}$	0.00 (0.00)	1.68	.093
Self-nudge	0.07 (0.04)$^{+}$	0.12 (0.07)	1.74	.083	0.06 (0.04)	0.09 (0.07)	1.34	.180
Transparency	0.07 (0.04)	0.11 (0.07)	1.52	.130	0.04 (0.04)	0.07 (0.07)	0.98	.327
Self-nudge × Transparency	0.00 (0.06)	0.00 (0.09)	0.02	.985	0.03 (0.06)	0.04 (0.08)	0.47	.638
Num.Obs.	399				397			
$_R2$	0.03				0.14			
R^2Adj	0.02				0.10			
F	3.57				3.51			

Note: $^{+}p < .1$, $p < .05$, $p < .01$, $p < .001$; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 12 includes control variables.

Appendix D. Instrumental variable regressions

To test whether the nudge effect of our digital salience nudge differed between the preset and the self-nudging condition, we calculated four additional instrumental variable regression models – two for the proportion of unhealthy food products and two for the proportion of healthy food products each. The models contained the following independent variables as predictors of the proportion of (un-) healthy food products in the participants' shopping baskets: *Nudge* (dummy that equals one for the preset nudge groups), *Transparency* (dummy that equals one for the transparency groups), *Self-nudge Yes* (dummy that equals one for participants in the self-nudging groups who chose to implement the nudge), and the respective interaction terms of the *Transparency* dummy with the *Nudge* and *Self-nudge Yes* dummies. To account for endogenous self-selection effects in the self-nudging groups, we added the *Self-nudge* variable (dummy that equals one for participants in the self-nudging groups) as an instrument to the regressions. For models 16 and 18, we further added the same control variables as in the main regressions reported in the main results section of this manuscript.

As the first stage of our instrumental variable regressions, we assessed whether our instrument (i.e., the random assignment to the self-nudging groups) is a significant predictor of the endogenous variable (i.e., the choice to implement the nudge). To do so, we calculated a linear regression model (see Model 13 in Table D1), including the independent endogenous variables *Self-nudge* and *Transparency* as predictors of the participant's decision to implement the nudge. The model was significant (F (2, 597) = 374.4, p <.001, $R^2 = 0.55$) and revealed a significant positive main effect of *Self-nudge*, indicating that the variable is an appropriate instrument. In addition, we calculated a second linear regression model (see Model 14 in Table D1), including the same predictors as the first model and the following endogenous control variables: the participant's gender, age, employment status, education level, current appetite, trait self-control, the strength of their healthy food choice motive. Again, the model was significant (F (16, 580) = 50.32, p <.001, $R^2 = 0.57$) and revealed a significant positive main effect of *Self-nudge*, supporting the results of the first model and indicating that the variable is an appropriate instrument.

D.1. *Instrumental variable regressions for the proportion of unhealthy food products*

The first model (see Model 15 in Table D2) was significant in predicting the proportion of unhealthy food products (F (5, 594) = 9.26, p < .001, $R^2 = 0.11$). It revealed significant negative main effects of *Nudge*, *Transparency*, *Self-nudge Yes*, and a significant positive interaction effect of *Nudge* × *Transparency*.

As we ran this regression model to test whether the nudge effects between a preset digital salience nudge and a self-selected digital salience nudge differed, we conducted a Wald-test-based comparison of the coefficients using the linearHypothesis function of the *car* package in R (Fox & Weisberg, 2019; R Core Team, 2023). The test did not reveal a significant difference ($\chi^2(1, 594) = 0.18$, p = .676), indicating that the effects of our digital salience nudge did not differ between the preset and the self-nudging groups. In addition, we conducted another Wald-test-based comparison, investigating whether the linear combination of the coefficients for *Nudge*, *Transparency*, and *Nudge*

× *Transparency* differs from the linear combination of the coefficients for *Self-nudge Yes* and *Transparency*, and *Self-nudge Yes* × *Transparency*. The test did not reveal a significant difference ($\chi^2(1, 594) = 1.12$, p = .289).

To test the robustness of these results, we ran a second regression model (see Model 16 in Table D2), that, additionally, included the following control variables: gender, age, employment status, education level, current appetite, trait self-control, strength of healthy food choice motive, and the total price of the participant's shopping basket. The second model was also significant in predicting the proportion of unhealthy food products (F (20, 576) = 8.15, p <.001, $R^2 = 0.23$). As our first model, it showed three significant main effects for *Nudge*, *Transparency*, and *Self-nudge Yes*. The interaction effect of *Nudge* × *Transparency* was marginally significant and the respective Wald-test-based comparisons remained insignificant (all p >.05), providing support for the results of our first model.

Table D1. First stage of the instrumental variable regression.

	Model 13				Model 14			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	−0.02 (0.02)	0.00 (0.00)	−1.03	.305	−0.16 (0.08)$^+$	0.00 (0.00)	−1.94	.053
Self-nudge	0.65 (0.02)	0.74 (0.03)	27.33	<.001	0.66 (0.02)	0.75 (0.03)	27.49	<.001
Transparency	0.04 (0.02)	0.04 (0.03)	1.62	.105	0.03 (0.02)	0.04 (0.03)	1.29	.197
Num.Obs.	600				597			
R^2	0.56				0.58			
R^2 *Adj*	0.55				0.57			
F	374.43				50.32			

Note: ^+p < .1, p < .05, p < .01, p < .001; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 14 includes control variables.

Table D2. Results of the IV regression models for the proportion of unhealthy food products selected by the participants.

	Model 15				Model 16			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.19 (0.02)	0.00 (0.00)	11.95	<.001	0.25 (0.05)	0.25 (0.05)	5.10	<.001
Nudge	−0.12 (0.02)	−0.33 (0.06)	−5.28	<.001	−0.11 (0.02)	−0.11 (0.02)	−5.07	<.001
Transparency	−0.07 (0.02)	−0.21 (0.07)	−3.07	.002	−0.06 (0.02)	−0.06 (0.02)	−2.64	.008
Self-nudge Yes	−0.10 (0.04)	−0.26 (0.09)	−2.81	.005	−0.10 (0.03)	−0.10 (0.03)	−2.82	.005
Nudge × Transparency	0.07 (0.03)	0.16 (0.07)	2.21	.028	0.05 (0.03)$^{+}$	0.05 (0.03)	1.79	.075
Self-nudge Yes × Transparency	0.03 (0.05)	0.05 (0.09)	0.54	.591	0.01 (0.05)	0.01 (0.05)	0.33	.743
Num.Obs.	600				597			
R^2	0.12				0.25			
$R^2 Adj$	0.11				0.23			
AIC	−500.4				−572.3			
BIC	−469.6				−475.7			
RMSE	0.16				0.14			

Note: $^{+}p < .1$, $p < .05$, $p < .01$, $p < .001$; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 16 includes control variables.

D.2. *Instrumental variable regressions for the proportion of healthy food products*

We also replicated the analysis outlined in E.1 for the proportion of healthy food products that represents our second key indicator for the healthiness of the participants' food product choices.

The first model (see Model 17 in Table D3) was significant in predicting the proportion of unhealthy food products (F (5, 594) = 3.34, p = .025, R^2 = 0.02). It revealed a significant positive main effect of *Nudge* and a marginally significant positive main effect of *Self-nudge Yes*.

To identify potential differences in the effectiveness of a preset digital salience nudge and a self-selected digital salience nudge differed, we conducted a Wald-test-based comparison of the coefficients. The test did not reveal a significant difference (χ^2(1, 594) = 0.07, p = .797), indicating that the effects of our digital salience nudge did not differ between the preset and the self-nudging groups. In addition, we conducted another Wald-test-based comparison, investigating whether the linear combination of the coefficients for *Nudge*, *Transparency*, and *Nudge* × *Transparency* differs from the linear combination of the coefficients for *Self-nudge Yes* and *Transparency*, and *Self- nudge Yes* × *Transparency*. The test did not reveal a significant difference (χ^2(1, 594)
= 1.22, p = .270).

Again, to test the robustness of these results, we ran a second regression model (see Model 18 in Table D3) including control variables. The second model was also significant in predicting the proportion of unhealthy food products (F (20, 576) = 4.18, p <.001, R^2 = 0.10). It revealed a significant positive main effect of *Nudge*, but no other significant main or interaction effects. As the coefficient for *Self-nudge Yes* was very close to being marginally significant, we conducted the same Wald-test-based comparisons of the models' coefficients that all remained insignificant (all p >.05).

Table D3. Results of the IV regression models for the proportion of healthy food products selected by the participants.

	Model 17				Model 18			
	B (SE)	*β (SE)*	*t*	*p*	*B (SE)*	*β (SE)*	*t*	*p*
Intercept	0.46 (0.03)	0.00 (0.00)	15.58	<.001	0.26 (0.09)	0.26 (0.09)	2.72	.007
Nudge	0.14 (0.04)	0.22 (0.07)	3.35	.001	0.13 (0.04)	0.13 (0.04)	3.10	.002
Transparency	0.07 (0.04)	0.11 (0.07)	1.56	.119	0.05 (0.04)	0.05 (0.04)	1.25	.210
Self-nudge Yes	0.12 (0.07)$^{+}$	0.17 (0.10)	1.79	.074	0.11 (0.07)	0.11 (0.07)	1.64	.101
Nudge × Transparency	−0.09 (0.06)	−0.11 (0.07)	−1.54	.123	−0.07 (0.06)	−0.07 (0.06)	−1.29	.199
Self-nudge Yes × Transparency	−0.02 (0.09)	−0.02 (0.10)	−0.19	.848	0.00 (0.09)	0.00 (0.09)	0.02	.986
Num.Obs.	600				597			
R^2	0.03				0.13			
$R^2 Adj$	0.02				0.10			
AIC	248.6				213.7			
BIC	279.4				310.3			
RMSE	0.29				0.28			

Note: $^{+}p < .1$, $p < .05$, $p < .01$, $p < .001$; *B* refers to unstandardised and *β* to standardised coefficients. Note: Model 18 includes control variables.